ADDIS ABABA UNIVERSITY
COLLEGE OF TECHNOLOGY AND BUILT ENVIRONMENT
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
APPLIED ELECTRONICS TWO PROJECT REPORT
SECTION 3A

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Design an op-amp based difference amplifier circuit shown in figure 5 with inputs V_1 and V_2 and output $V_0 = -5V_1 + 3V_2$. Additionally perform a simulation in proteus software to validate the obtained results.

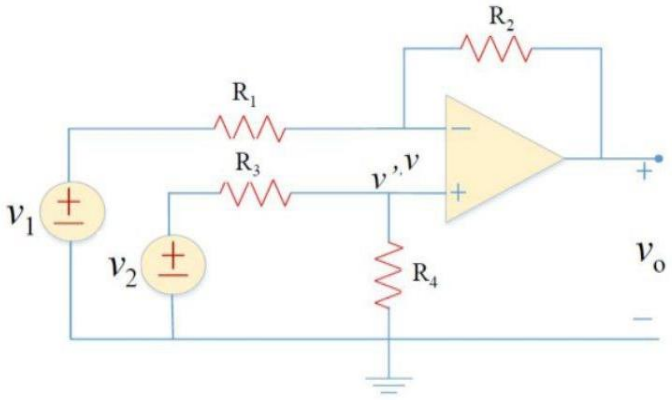


Fig. 5

$V_0 = -5V_1 + 3V_2$

at node v

$$\frac{V_1 - V}{R_1} = \frac{V - V_0}{R_2}$$

$$R_2(V_1 - V) = R_1(V - V_0)$$

$$R_2V_1 - R_2V = R_1V - R_1V_0$$

$$R_1V_0 = (R_1 + R_2)V - R_2V_1$$

$$V_0 = \frac{R_1 + R_2}{R_1}V - \frac{R_2}{R_1} \quad \text{.....(*)}$$

At node V`

$$\frac{V_2 - V`}{R_3} = \frac{V`}{R_4}$$

$$R_3V` = R_4(V_2 - V`)$$

$$R_3V` = R_4V_2 - R_4V`$$

$$R_3V` + R_4V` = R_4V_2$$

$(R_3 + R_4)V` = R_4V_2$

$$V` = \frac{R_4}{R_4 + R_3}V_2 \text{ .....(**)}$$

V = V`; substitute (**) into (*)

$$V_0 = \frac{R_1 + R_2}{R_1}\left(\frac{R_4}{R_4 + R_3)}V_2 - \frac{R_2}{R_1}V_1\right.$$

- $V_0 = -5V_1 + 3V_2$
- $-5V_1 = -\frac{R_2}{R_1}V_1$

$$5 = \frac{R_2}{R_1}$$

$R_2 = 5R_1$

$$\left(\frac{R_1 + R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)V_2 = 3V_2$$

$$\left(\frac{R_1 + R_2}{R_1}\right)\left(\frac{R_4}{R_4 + R_3}\right) = 3$$

$$\frac{R_1 + 5R_1}{R_1}\left(\frac{R_4}{R_3 + R_4}\right) = 3$$

$$\frac{6R_1}{R_1}\left(\frac{R_4}{R_4 + R_3}\right) = 3$$

$$6\frac{R_4}{R_4 + R_3} = 3$$

$$\frac{R_4}{R_3 + R_4} = \frac{3}{6}$$

$$\frac{R_4}{R_3 + R_4} = \frac{1}{2}$$

$$2R_4 = R_3 + R_4$$

$$2R_4 - R_4 = R_3$$

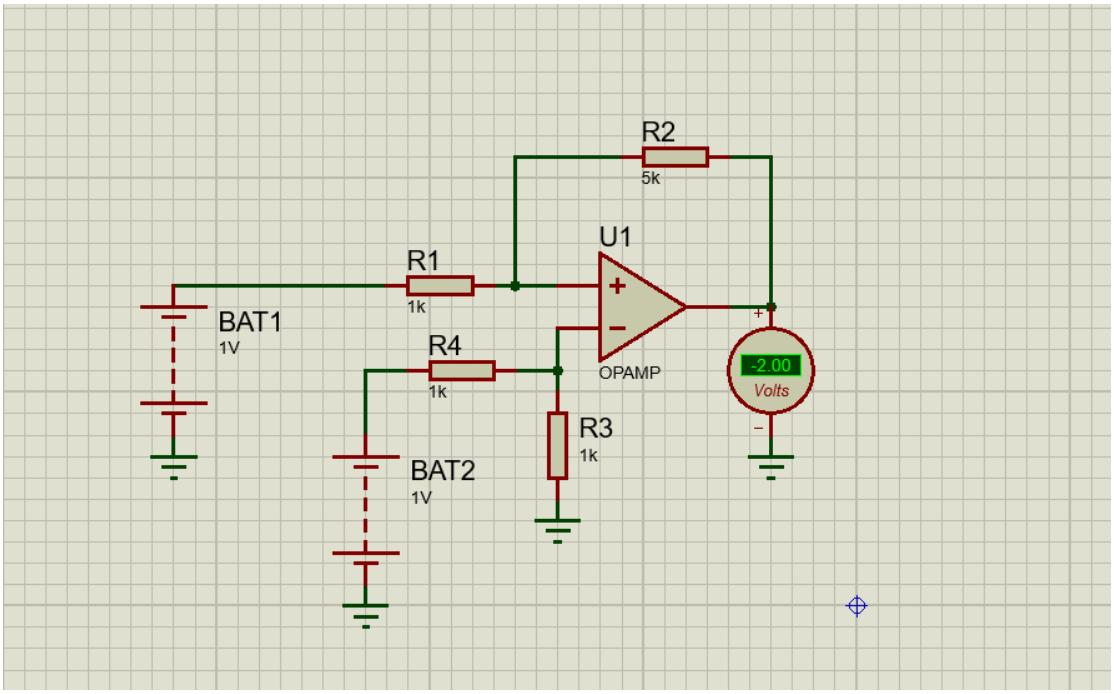
$R_4 = R_3$ $\qquad$ $R_2 = 5R_1$

Validating with proteus:

- We used the following resistor and input voltage in the simulation:
  - R4 = 1000ohm
  - R3 = 1000ohm
  - R2 = 5000ohm
  - R1 = 1000ohm
  - v1 = 1v
  - v2 = 1v
- Expected output:

  V(out) = -5(1v) + 3(1v)

  = -2v
- The following figure shows the connection:



- As showed form the figure the output voltage is -2v. Which proves that our calculation was correct.